Predictive Analytics Poster Rubric (Poster-Only Assessment)

#	Criterion	Points	Exemplary (Full Points)	Competent (Partial Points)	Needs Improvement (Minimal Points)
1	Prediction Goal(s)	5	5 pts: Clear, specific predictive objective(s); response type (regression/classification) explicitly identified and justified; strong business/research rationale.	3–4 pts: Goal stated; response type identified but weakly justified; limited link to context.	0–2 pts: Goal unclear/missing; response type unspecified; no rationale.
2	Dataset Overview	8	7–8 pts: Reports #obs/#features; types for key features; response variable identified/typed; data provenance/window stated.	4–6 pts: Size and response identified; partial feature typing; minimal provenance.	0–3 pts: Missing size/response ID; little/no feature typing or context.
3	Exploratory Data Analysis (EDA)	12	10–12 pts: Appropriate summaries (response + key predictors); ≥2 relevant plots; interprets skew/outliers/imbalance/seasonality; connects insights to modeling choices and metric selection.	6–9 pts: Basic stats and ≥1 plot; some interpretation but limited linkage to modeling.	0–5 pts: Incomplete stats; weak/irrelevant visuals; minimal/no interpretation.
4	Data Preparation (Feature Engineering, Missing Data, etc.)	18	15–18 pts: ≥1 non-trivial new feature (interactions, lags/rolling stats, domain-driven ratios, text-derived signals) with clear rationale; quantifies missingness by feature; justified strategy (drop/impute/advanced) with leakage-safe implementation; demonstrates impact via ablation/CV deltas or sensitivity to imputation choices.	9–14 pts: New feature with partial rationale; identifies missingness and applies a reasonable method; limited evidence of impact or sensitivity analysis.	0–8 pts: No new/trivial feature; missingness unreported or mishandled; unjustified/inappropriate method; no evidence of value.
5	Baseline Model (Choice, CV, Interpretation)	17	14–17 pts: Appropriate simple baseline (e.g., Linear/Logistic Regression); pipeline is leakage-safe; k=5/10 CV with correct metric (RMSE/MAE/Accuracy/F1/AUC); tracks mean/variance; summarizes implications and limitations; proposes concrete next steps.	8–13 pts: Reasonable model; CV present with partial integration/details; metric adequate; interpretation limited or generic.	0–7 pts: Inappropriate/unspecified baseline; no/incorrect CV; unclear metric; raw numbers only; no reflection.

#	Criterion	Points	Exemplary (Full Points)	Competent (Partial Points)	Needs Improvement (Minimal Points)
6	More Complex Model (Choice, Tuning, Paired Comparison)	25	22–25 pts: Model aligned to data/goal (RF/GBM/XGBoost/SVM/regularized GLM/NN); hyperparameter search (grid/random/Bayesian) inside proper CV; leakage controls (scaling/encoding inside folds); compared with the baseline on identical folds (same splitter and seed) using per-fold paired differences, signed so a positive value favors the complex model, against a practical margin stated before the results; mean paired difference reported with an approximate 95% interval corrected for fold overlap (mean ± t × SD of the differences × √(1/k + 1/(k−1))); verdict names the correct outcome: worthwhile gain (the whole interval lies above the margin: adopt the complex model), worthwhile loss (the whole interval lies below minus the margin: keep the baseline), practically equivalent (the whole interval lies inside ± the margin, so the simpler model wins), or unresolved (anything else: the simpler model is kept as a stated policy, not called equivalent); fold-by-fold wins reported as description, not as the verdict; practical interpretation (effect sizes, trade-offs), plus error/confusion analysis where relevant.	13–21 pts: Plausible model; some tuning with partial documentation; CV present but thin; compares with the baseline on CV means or side-by-side intervals rather than paired per-fold differences; judges a winner or a tie by whether intervals overlap or by fold signs alone; calls the models equivalent when the interval extends past the margin; or states no margin; limited practical takeaways.	0–12 pts: Misaligned/unspecified model; superficial/absent tuning; incorrect CV; opaque selection; no clear comparison or interpretation.
7	Report Quality & Visual Communication	10	9–10 pts: Clear, scannable poster; logical flow (Intro → Methods → Results → Conclusions); legible, labeled figures/tables (titles, axes, units, legends); consistent typography/layout; code refs/appendix where appropriate; minimal grammar issues.	6–8 pts: Generally readable; minor layout/labeling gaps; some figures under-explained or styling inconsistencies.	0–5 pts: Disorganized; illegible/unlabeled visuals; weak linkage between text and graphics; notable grammar/formatting issues.
8	Sources, Acknowledgments, AI Use & Data Privacy	5	5 pts: Data source cited (provider, URL or DOI, version or access date); references support the business-problem claims and the main software is credited (e.g., scikit-learn, pandas); Acknowledgments section names the Mitch Daniels School of Business, Prof. Moreira, the TAs, and other contributors; AI Use Disclosure names each AI tool and what it was used for and states that the team verified the output (or states that no AI tools were used); only aggregated or anonymized data and examples.	3–4 pts: AI Use Disclosure present but vague (names tools without saying what they did), or one other element missing or incomplete (e.g., acknowledgments omit the TAs; data source cited without references or tools). No privacy issue.	0–2 pts: No AI Use Disclosure, or two or more other elements missing. Any personally identifiable or NDA-restricted data on the poster scores 0 here and must be removed before printing.

Total: 100 points. This rubric grades M09 (poster first draft), M10 (final poster), and the Instructor/TA evaluation after the conference.

Criterion 6 at M09: a comparison that lacks per-fold paired differences, a stated margin, or the corrected interval gets feedback, not a deduction, on the first draft. All three are scored from M10.

Fall 2026: a comparison that follows notebook 08 as taught (the descriptive interval, and a verdict from the three outcomes: every fold favors the same model and the gap clears the margin; the sign changes across folds; or the same sign but under the margin) also earns full credit for the interval and the verdict, provided the outcome is applied correctly.

AI use: per the syllabus AI Policy, every poster carries an AI Use Disclosure (the template has a section for it). Errors in AI-assisted work are the team's responsibility.

Honors: a labeled honors section, where present, is graded only under the Honors Research Extension and is not scored under this rubric.